Table 1: Clustering quality results for the SWAT dataset.

				Lloyd		Cost			$\text{Gap}^{\text{SFC}}_{\text{Cost}}$ [%]		Balance			
k	Target	p	q	Cost	Balance	S-MPFC	Flow	SFC	S-MPFC	Flow	Lloyd	S-MPFC	Flow	SFC
2	0.06	6	100	1,207,610	0.02	1,381,431	**1,367,196**	1,384,669	-0.23	-1.26	✗	✓	✓	✓
2	0.04	4	100			1,303,876	**1,273,623**	1,376,146	-5.25	-7.45	✗	✓	✓	✓
2	0.02	1	50			1,207,940	**1,207,610**	1,316,615	-8.25	-8.28	✓	✓	✓	✓
5	0.06	6	100	556,811	0.02	1,027,589	**995,870**	1,066,735	-3.67	-6.64	✗	✓	✓	✓
5	0.04	4	100			847,464	**772,824**	1,008,520	-15.97	-23.37	✗	✓	✓	✓
5	0.02	1	50			557,388	**557,213**	980,700	-43.16	-43.18	✓	✓	✓	✓
10	0.06	6	100	256,437	0.00	871,850	**768,414**	823,198	5.91	-6.66	✗	✓	✓	✓
10	0.04	4	100			735,147	**477,826**	865,644	-15.08	-44.80	✗	✓	✓	✓
10	0.02	1	50			302,460	**260,686**	761,262	-60.27	-65.76	✗	✓	✓	✓
15	0.06	6	100	196,223	0.00	855,302	**711,829**	772,028	10.79	-7.80	✗	✓	✓	✓
15	0.04	4	100			716,984	**421,615**	744,505	-3.70	-43.37	✗	✓	✓	✓
15	0.02	1	50			277,073	**203,142**	611,812	-54.71	-66.80	✗	✓	✓	✓
Average									-16.13	-27.11				

(**bold**) best result; (underline) 2nd best result; (✓) target balance attained; (✗) target balance not attained

Table 2: Running times for the SWAT dataset.

				CPU_{tot} [s]			CPU_{avg} [s]			$\text{Gap}^{\text{SFC}}_{\text{CPU}_{\text{avg}}}$ [%]		Avg. iterations	
k	Target	p	q	S-MPFC	Flow	SFC	S-MPFC†	Flow	SFC	S-MPFC	Flow	S-MPFC	Flow
2	0.06	6	100	10.51	1,000.29	limit	1.09	10.00	856.69	-99.87	-98.83	1.95	2.27
2	0.04	4	100	10.84	1,071.64	limit	1.10	10.72	817.71	-99.87	-98.69	1.78	2.53
2	0.02	1	50	6.43	974.94	limit	1.05	9.75	590.50	-99.82	-98.35	1.55	2.20
5	0.06	6	100	75.92	1,409.81	limit	1.75	14.10	825.44	-99.79	-98.29	2.97	2.28
5	0.04	4	100	32.32	1,408.57	limit	1.31	14.09	834.77	-99.84	-98.31	1.98	2.28
5	0.02	1	50	14.85	1,375.23	limit	1.14	13.75	575.08	-99.80	-97.61	2.00	2.23
10	0.06	6	100	3,453.49	2,217.43	limit	39.70	22.17	828.17	-95.09	-97.32	2.84	2.53
10	0.04	4	100	107.07	2,229.30	limit	2.06	22.29	802.24	-99.74	-97.22	2.10	2.57
10	0.02	1	50	58.09	2,254.66	limit	1.57	22.55	587.53	-99.73	-96.16	2.13	2.65
15	0.06	6	100	1,714.40	3,004.45	limit	142.87	30.04	827.22	-82.61	-96.37	3.50	2.63
15	0.04	4	100	261.39	3,182.92	limit	3.60	31.83	804.15	-99.55	-96.04	2.23	2.93
15	0.02	1	50	110.66	3,246.30	limit	2.10	32.46	588.87	-99.64	-94.49	2.40	3.08
Average										-97.95	-97.31	2.29	2.52

† including preprocessing time; (limit) time limit of 3,600 seconds reached

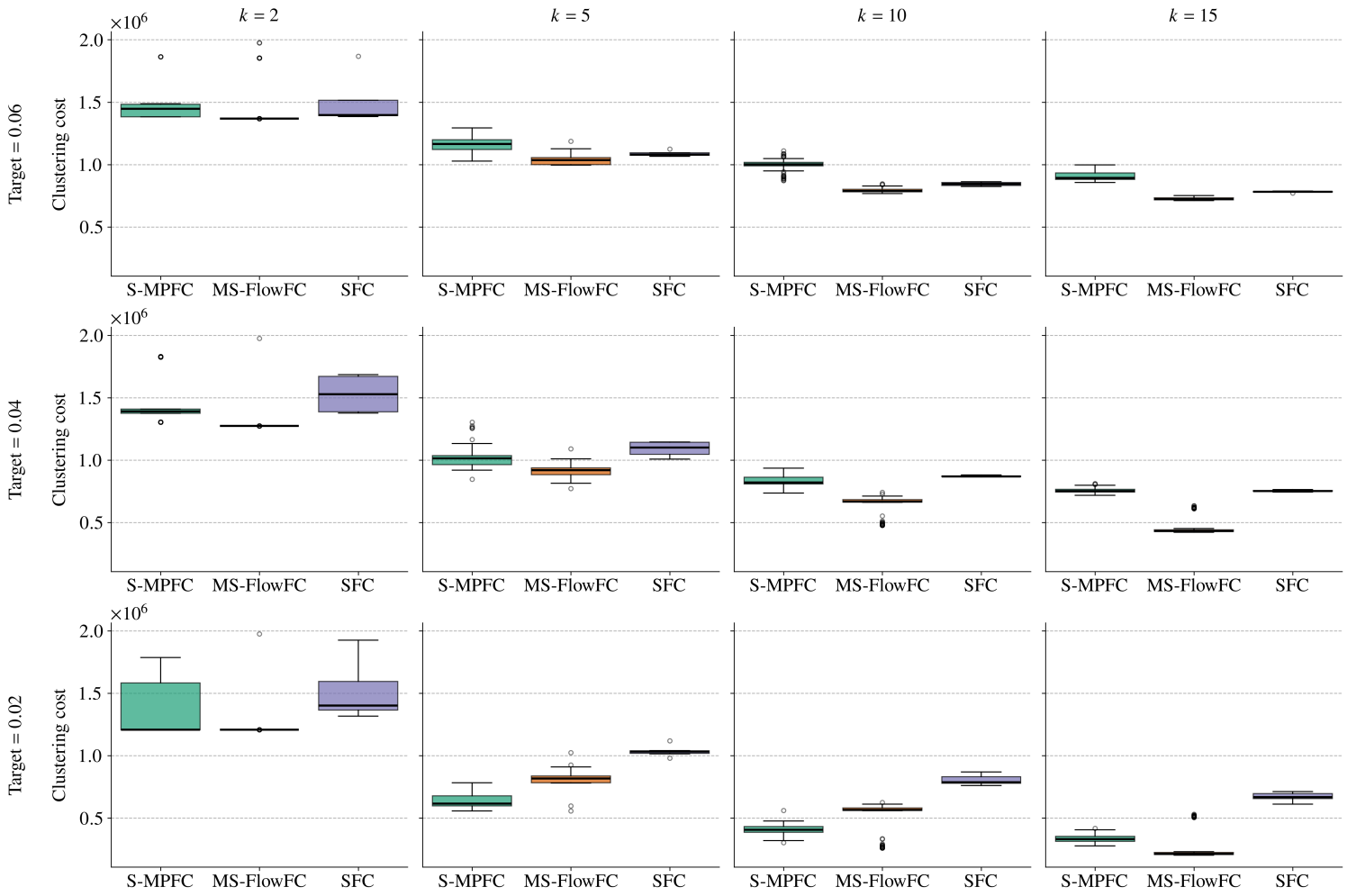

Figure 1: Boxplots of clustering costs for solutions obtained with different random seeds on the SWAT dataset, across different methods, numbers of clusters, and tolerance values.